

Reliability-Aware Structured Synthetic Data Generation via Schema Enforcement and Layered Repair

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Abstract

Large language models (LLMs) are increasingly used to generate structured synthetic datasets, yet raw LLM outputs frequently violate schema constraints, produce malformed JSON, and exhibit semantic drift under retrieval and multimodal conditioning. We present a controlled empirical study of structured synthetic data generation reliability across four experimental conditions using a locally deployable pipeline built on LLaMA3 8B. Our pipeline combines schema enforcement, a multi-layer post-hoc repair loop, retrieval-augmented grounding (embedding-based and TF-IDF), and optional vision-language model (VLM) conditioning via LLaVA 7B. We make three concrete contributions: (1) a quantitative demonstration that schema enforcement raises JSON validity from 70% to 94–98% on a 100-query benchmark without model fine-tuning, a +24 to +28 percentage point lift; (2) a failure taxonomy from the repair loop showing that complex nested schemas trigger 25 repair activations across 30 queries, recovering 73.3% of initially invalid outputs with a clear breakdown between JSON parse errors and schema validation errors; and (3) an empirical measurement of semantic alignment degradation under retrieval grounding ($\Delta = -0.039$ for TF-IDF) and multimodal conditioning ($\Delta = -0.006$ for matched VLM pairs). All experiments run on a consumer-grade RTX 3050 6 GB GPU, establishing a reproducible baseline for resource-constrained synthetic data pipelines.

CCS Concepts

• **Information systems** → **Data management systems**; *Data cleaning*; • **Computing methodologies** → **Natural language processing**; *Artificial intelligence*; *Vision and language*.

Keywords

Synthetic data generation, structured generation, schema enforcement, post-hoc-repair JSON repair, retrieval-augmented generation, multimodal grounding, LLM reliability, resource-constrained deployment

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1 Introduction

The demand for large-scale, high-quality structured datasets has accelerated with the adoption of data-centric AI [10, 23]. Synthetic data generation using large language models offers a promising path to filling data gaps in scarce or sensitive domains [13]. However, while LLMs excel at generating unstructured text, their ability to maintain high-fidelity labels in low-resource settings remains a critical bottleneck [16]. Structured JSON generation with LLMs presents a critical reliability problem [15]. When given a prescribed schema, models often violate field constraints. They produce malformed syntax or hallucinate attributes absent from the query context. This challenge remains systematically underexplored.

While foundational research demonstrates that these models can act as highly effective few-shot learners [3], structured output generation is a different challenge. Schema adherence is highly sensitive to prompt design and instructional overhead. Chain-of-Thought research [20] shows this clearly: structured reasoning is an emergent behavior, not an inherent model trait. Without explicit guidance, LLMs deviate from prescribed relational schemas.

Existing approaches to structured generation fall into two broad categories. *Constrained decoding* methods [1, 14, 21] enforce schema compliance at the token-sampling level. It guarantees valid output but requires white-box model access. This is unavailable in local inference deployments such as Ollama. *Post-hoc repair* detects and corrects invalid outputs after generation. It is more deployment-flexible but lacks systematic study of failure modes and recovery rates. We address three research questions:

- **RQ1.** How much does multi-layer post-hoc schema enforcement improve JSON validity over unguided LLM generation on structured queries?
- **RQ2.** Under what schema complexity conditions does the repair loop engage, and what is its recovery rate?
- **RQ3.** How do retrieval grounding and VLM-based multimodal conditioning affect semantic alignment in structured outputs?

Our key results are as follows. Schema enforcement raises JSON validity by +24 to +28 percentage points over unguided generation (RQ1). On complex nested schemas, the repair loop achieves a 73.3% recovery rate with a clear two-layer failure taxonomy (RQ2). Both retrieval conditions introduce consistent but small alignment degradation (RQ3). All experiments run on a consumer 6 GB GPU via a fully local pipeline. No cloud infrastructure is required.

Figure 1 provides a visual overview of the approach and key results across the three experimental conditions. The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the pipeline architecture. The experimental setup is described in Section 4. Section 5 then presents results.



Figure 1: Overview of our approach. Without schema enforcement (left), LLaMA3 8B produces valid structured JSON on only 70% of queries, with failures split between malformed syntax and type violations. Our multi-layer repair pipeline (center) applies sequential syntactic and schema-level correction under strict additionalProperties: false validation, raising validity to 94–98% without model fine-tuning (right). All experiments run on a consumer RTX 3050 6 GB GPU.

Section 6 discusses implications and limitations. Finally, Section 7 offers the concluding remarks.

2 Related Work

Structured generation and constrained decoding. Outlines [21] and Guidance [14] ensure adherence to schemas through finite-state machine-driven token sampling. Valid output is guaranteed. However, white-box model access is required. Beyond simple regex-based constraints, grammar-constrained decoding extends this further [5]. It handles structured NLP tasks without fine-tuning. This research direction addresses the inherent ‘asymmetry’ in synthetic data generation, where the complexity of the target relational schema often exceeds the model’s zero-shot structural priors [7].

LMQL [1] takes a different route. It was built upon this foundation by introducing a declarative query language designed for constrained decoding. OpenAI function calling [17] and the Instructor library [12] provide schema enforcement at the API level. Our work targets *black-box local inference* (Ollama). Here the token-level constraints are unavailable, making post-hoc repair the only viable enforcement mechanism. This deployment scenario is practically important for privacy-sensitive and resource-constrained settings.

Retrieval-augmented generation. RAG systems [9] improve factual grounding by injecting retrieved context into the prompt. Recent work has shown that retrieval noise—irrelevant or low-quality context—can degrade generation quality [19]. Our work extends this observation specifically to *structured output quality*, showing that retrieval noise measurably reduces semantic alignment scores.

Multimodal grounding in generation. LLaVA [11] and related vision-language models enable image-conditioned text generation. Prior work has studied VLM grounding for open-ended generation; however, the effect of VLM conditioning on *structured output*

alignment when query-image relevance varies has not been systematically measured. We provide the first controlled measurement of VLM-induced alignment drift in a structured generation pipeline under matched image-query conditions.

Synthetic data quality evaluation. Prior work on synthetic data evaluation focuses on fidelity to real distributions [6] and downstream utility [2]. Recent work has specifically examined the potential and limitations of LLM-generated synthetic data for text classification tasks, finding that while prompted generation can reduce labeling overhead, structural reliability degrades as task complexity increases [10]. Structural validity—whether generated outputs align with established schemas—has not been subjected to rigorous scrutiny. Our evaluation framework and failure taxonomy directly contribute to this gap.

3 System Architecture

Figure 2 illustrates the pipeline. The system is organized into five stages: query input, retrieval grounding, LLM generation, multi-layer repair, and evaluation. The pipeline is designed around a single operational principle: every component that adds context to the generation prompt (retrieval, VLM conditioning, HyDE expansion) operates upstream of schema enforcement, ensuring that semantic enrichment never bypasses structural validation.

3.1 Retrieval Module

Two retrieval backends are supported to provide a robust comparison between lexical and semantic grounding. **Embedding retrieval** uses BAAI/bge-small-en-v1.5 [22] to embed queries and a pre-built document corpus, returning top- k passages by cosine similarity above a configurable threshold τ . Our choice of a dense vector backend is grounded in the success of **Dense Passage Retrieval**

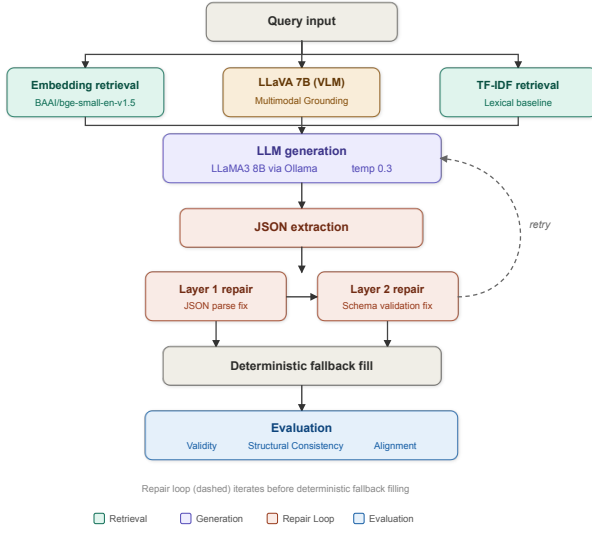


Figure 2: Pipeline architecture. Query input feeds two parallel retrieval backends (embedding and TF-IDF) and optional VLM conditioning. The repair loop (dashed arrow) iterates between Layer 1 JSON parse repair and Layer 2 schema validation repair before deterministic fallback filling.

(DPR) [8], which established that semantic embeddings significantly outperform traditional sparse methods for open-domain context identification. In contrast, **TF-IDF retrieval** uses BM25-style term matching via scikit-learn, serving as a lexical baseline. Both backends are ablation-ready: setting $k = 0$ disables retrieval entirely (Stage 0 baseline).

Prior to retrieval, all stages apply HyDE query expansion [4]: the LLM generates a hypothetical ideal structured record from the raw query, and the resulting document embedding serves as the retrieval vector. This bridges the lexical gap between short natural-language queries and schema-relevant corpus documents.

3.2 VLM Conditioning

VLM grounding builds on vision-language alignment research [18]. The core principle is contrastive natural language supervision. We adapt this to structured generation. Visual features are aligned with textual descriptions. This grounds generated JSON in multimodal evidence rather than LLM-internal hallucinations. In our implementation, image input triggers LLaVA 7B [11] via Ollama. It produces a natural-language caption of the image. This caption is appended to the generation prompt as grounding context. For Stage 2, queries are matched to images by semantic relevance (Section 4).

3.3 Multi-Layer Repair Loop

Algorithm 1 defines the recovery procedure. Error handling is split into two layers. This avoids unnecessary LLM inference calls.

Layer 1 (Syntactic Repair) targets JSONDecodeError events. It applies heuristic fixes: bracket balancing and quote normalization. **Layer 2 (Schema Repair)** targets structural non-compliance. The

Input: LLM output o , schema S , max attempts K

Output: Valid JSON v , repair trace T

$T \leftarrow []$;

$i \leftarrow 1$;

while $i \leq K$ **do**

if $\text{is_valid_json}(o)$ **then**

$v \leftarrow \text{parse}(o)$;

if $\text{validate}(v, S)$ **then**

return v, T ;

else

$e \leftarrow \text{schema_error}(v, S)$;

$T.\text{append}(\{\text{layer} : \text{schema}, o, e\})$;

$o \leftarrow \text{repair_schema}(o, S, e)$;

end

else

$e \leftarrow \text{parse_error}(o)$;

$T.\text{append}(\{\text{layer} : \text{json_parse}, o, e\})$;

$o \leftarrow \text{repair_json}(o)$;

end

$i \leftarrow i + 1$;

end

return $\text{enforce_minimum}(o, S), T$;

Algorithm 1: Multi-Layer Schema Repair

Draft7Validator detects type mismatches, missing fields, and nesting violations. Each is corrected via targeted re-prompting.

The validator runs in **strict mode** (additionalProperties: false). Any extra field from the LLM triggers a validation failure. This includes conversational metadata or unrequested context. The design trades repair count for output cleanliness. Generated data strictly follows the target schema.

When both layers exhaust the retry budget, **deterministic fallback filling** activates. It inserts schema-compliant defaults into missing or invalid fields. Every intervention is logged to a repair_trace object. Each entry records the layer, the raw LLM response, and the validation error. This enables downstream failure taxonomy analysis.

Repair Latency: Each repair attempt incurs one additional LLM inference call. On the RTX 3050 6 GB hardware, a single LLaMA3 8B generation takes approximately 8–12 seconds. The repair loop therefore adds up to $2K$ additional calls per query in the worst case. For Stage 1, the 72 total repairs across 100 queries represent an average overhead of 0.72 additional calls per query.

The five stages are designed to be independently ablatable: any component (retrieval backend, VLM conditioning, repair loop) can be disabled via parameter flags, enabling the controlled experimental comparisons described in Section 4.

3.4 Evaluation Module

Three metrics are computed per query:

JSON Validity Rate is the fraction of outputs that parse successfully and pass schema validation after the repair loop:

$$\text{VR} = \frac{|\{o_i : \text{valid}(o_i)\}|}{N}.$$

Structural Consistency Score¹ measures the fraction of required schema fields present and correctly typed in the output:

$$SC = \frac{1}{N} \sum_{i=1}^N \frac{|\text{fields_correct}(o_i)|}{|\text{fields_required}(S)|}.$$

Query-Object Alignment Score measures semantic similarity between the query and the generated object descriptions using BAAI/bge-small-en-v1.5 embeddings:

$$QA = \frac{1}{N} \sum_{i=1}^N \cos(\mathbf{e}_{q_i}, \mathbf{e}_{o_i}),$$

where $\mathbf{e}_{q_i}$ and $\mathbf{e}_{o_i}$ are the query and output embeddings, respectively.

Embedding cosine similarity measures semantic proximity between query and output. It does not verify factual correctness or downstream task utility. Evaluation against task-specific performance metrics remains future work.

4 Experimental Setup

4.1 Model and Hardware

All experiments use LLaMA3 8B served via Ollama (temperature 0.3, num_predict 256) on a single NVIDIA RTX 3050 6 GB GPU with 16 GB system RAM. No model fine-tuning is performed. VLM experiments use LLaVA 7B via the same Ollama backend.

4.2 Dataset

The **standard benchmark** consists of 100 structured scene-description queries. The queries are drawn from `raw_queries.json` covering objects, actions, spatial relationships, and attributes. Queries are short natural-language descriptions (e.g., “a person riding a bicycle near a park”). The target schema is a two-level JSON object with required fields `scene_description` (string), `objects` (array of strings), and `actions` (array of strings).

The **complex nested schema** benchmark uses a three-level schema structure. It contains arrays of typed objects, numeric constraints, and inter-field dependencies. The schema requires a top-level scene object containing an `entities` array. Each entity carries typed properties and a nested `spatial_relations` array with numeric distance fields. Thirty queries target this schema. Both JSON parse repair and schema validation repair are expected to activate on this benchmark.

For **Stage 2**, 15 matched image-query pairs are constructed: five queries per image (`dog.jpg`, `car.jpg`, `football.jpg`), where each query semantically describes the image content.

4.3 Experimental Conditions

We evaluate the pipeline across four incremental conditions as shown in Table 1. To bridge the semantic gap between short user queries and structured schema requirements, all stages utilize **HyDE query expansion** [4]. This ensures a high-quality retrieval context. In this configuration, the LLM first generates a hypothetical “ideal” synthetic record based on the user query, which is then used as the search vector for the retrieval backends.

¹The current implementation scores objects and actions presence; `scene_description` is validated by the schema validator separately.

Table 1: Experimental conditions across four pipeline stages. All configurations utilize HyDE for query expansion. N denotes the number of queries processed per stage.

Stage	Retrieval	HyDE	VLM	Enforce	N
0 – Baseline	None	✓	×	×	100
1 – Embedding	BGE-Small	✓	×	✓	100
2 – VLM Grounded	BGE-Small	✓	✓	✓	15
3 – TF-IDF	TF-IDF	✓	×	✓	100

The sample distribution is bifurcated by modality: text-only retrieval (Stages 0, 1, and 3) utilizes a larger corpus to ensure statistically robust validity metrics, while the multimodal configuration (Stage 2) focuses on a high-fidelity “pilot” set of matched image-query pairs to evaluate vision-language alignment. The configuration of Stage 2 (VLM Grounded) follows a conditional trigger logic: image assignment via `get_image_for_query()` is performed through keyword matching on the input text. Consequently, only the 15 queries containing the target keywords (*dog*, *car*, *football*) trigger the LLaVA-7B VLM conditioning. For queries outside this keyword set, the stage effectively reverts to text-only embedding retrieval, ensuring the $N = 15$ matched pairs serve as a controlled measurement of vision-language alignment without introducing noise from ungrounded queries.

5 Results

5.1 RQ1: Effect of Schema Enforcement on Validity

Table 2 presents our primary performance metrics. The baseline results (Stage 0) show that LLaMA3 8B only makes JSON that follows the schema in **70%** of cases when there is no enforcement. A qualitative audit of the 30 baseline failures shows a mix of syntactic and semantic problems. There are 21 instances of malformed JSON structures (`malformed_json`) and 9 instances of schema-constraint violations (`wrong_type`). This confirms that both parse-level and type-level errors are systemic at this model scale.

Adding schema enforcement with embedding retrieval (Stage 1) increases validity to **98%**—a **+28 percentage point** improvement. TF-IDF retrieval with enforcement (Stage 3) achieves **94%**—a **+24 pp** improvement. Most importantly, enforcement completely eliminates schema-level type errors, which are fully resolved by the enforcement mechanism while a small residual of parse-level failures persists. Figure 3 shows this validity gap across all three stages.

Computational Overhead and Repair Dynamics. The repair counts in Table 2 show the total number of loop activations for all queries. For Stage 1, the 72 total repairs show that queries that failed needed an average of 2.4 more LLM calls to be valid. The zero-repair result in Stage 2 is notable. The VLM caption provided dense, entity-level context. This appears to have guided the LLM toward schema-compliant output in a single pass, without requiring repair loop activation. Preliminary evidence suggests that detailed image captions may act as an implicit schema scaffold. Specifically, grounding concrete visual entities in the prompt may reduce structurally underspecified outputs before validation. This observation is

Table 2: Main Results: standard schema, LLaMA3 8B via Ollama. Alignment and structural scores are not computed for Stage 0 as outputs are not schema-aligned. Stage 0 serves as the baseline. Repairs are the total number of LLM retry activations for the whole test set ($N = 100$ for Stages 1 and 3; $N = 15$ for Stage 2). Individual queries may trigger multiple repairs up to the $K = 2$ limit per layer.

Stage	Valid%	Struct.	Align.	Total Repairs
0 – No enforcement	70.0	—	—	—
1 – Embedding RAG	98.0	0.725	0.938	72
2 – Embedding+VLM	98.0	0.700	0.932	0
3 – TF-IDF RAG	94.0	0.700	0.899	53

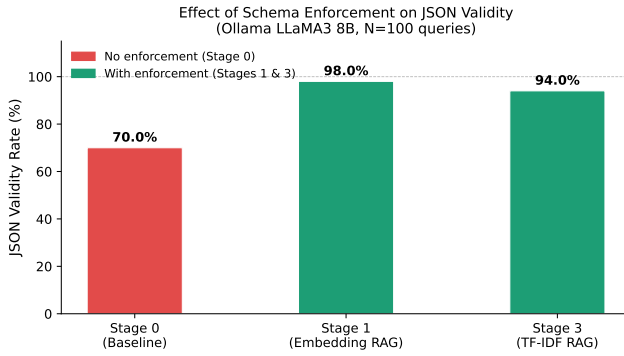


Figure 3: JSON validity rate by stage. Stage 0 (no enforcement) achieves 70%; schema enforcement raises validity to 94–98%.

Table 3: Repair loop results: complex nested schema (3-level nesting, array constraints, $N=30$).

Metric	Value
Total queries	30
Valid after repair	22 (73.3%)
Unrecoverable	8 (26.7%)
JSON parse repairs	18
Schema validation repairs	7
Total repair attempts	25

based on a small pilot ($N=15$) and should be interpreted cautiously. Confirmation at larger scale is a direct priority for future work.

5.2 RQ2: Repair Loop Behavior Under Complex Schema

Table 3 presents the complex nested schema results. The repair loop activates on **83%** of queries. Total repair attempts reach 25 across 30 queries. JSON parse errors drive 72% of activations (18/25). Schema validation errors account for the remaining 28% (7/25). These rates reflect a known pattern. LLM structured generation has a reliability ceiling under increasing schema complexity [10]. Deeper nesting and tighter inter-field constraints push models past this ceiling.

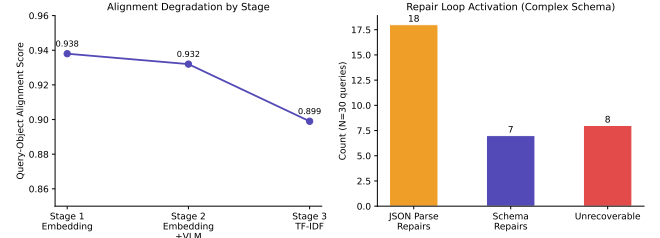


Figure 4: Left: query-object alignment degradation across stages. Right: repair loop activation breakdown for complex schema ($N=30$).

Even well-prompted models then require iterative correction to reach schema compliance.

After repair, **22/30 queries** (73.3%) produce valid output. Eight queries (26.7%) remain unrecoverable after K repair attempts and fall through to deterministic minimum-field enforcement. This result demonstrates two things simultaneously: the repair loop provides meaningful recovery—restoring nearly three-quarters of initially invalid complex-schema outputs—while also revealing the limits of post-hoc repair under deep structural complexity. The 26.7% unrecoverable rate characterizes the regime where constrained decoding methods would be preferable.

5.3 RQ3: Alignment Degradation Under Grounding

Comparing alignment scores across Stages 1–3 reveals a consistent degradation pattern. Figure 4 (left panel) visualises the alignment score trajectory across the three enforced stages, while the right panel shows the repair activation breakdown for the complex schema experiment described in Section 5.2. Embedding-only retrieval (Stage 1) achieves the highest alignment at 0.938. Adding matched VLM conditioning (Stage 2) reduces alignment by $\Delta = -0.006$ to 0.932. TF-IDF retrieval (Stage 3) shows the largest degradation at $\Delta = -0.039$, reaching 0.899. Structural consistency follows the same trend: Stage 1 scores 0.725 while Stages 2 and 3 both score 0.700.

We interpret these findings as follows. First, the VLM drop ($\Delta = -0.006$) is small but consistent across all 15 matched pairs. It suggests that even semantically relevant image captions introduce distributional shifts in the generation prompt. This marginally reduces query-output alignment. Second, the TF-IDF drop ($\Delta = -0.039$) is larger. Lexical matching retrieves topically relevant but semantically imprecise context [19]. This dilutes prompt focus. Both effects also appear in structural consistency scores. Context pollution degrades schema adherence alongside semantic alignment.

6 Discussion

Post-hoc repair as an alternative to constrained decoding. Our results show that post-hoc repair achieves 94–98% validity on standard schemas. No white-box model access is required. This makes it a practical alternative to constrained decoding (Outlines, Guidance) in local deployment scenarios. The local deployment systems do not allow token-level intervention. The +28 pp lift over unguided

generation (70% $\rightarrow$ 98%) indicates that the baseline failure rate is substantial enough to justify the repair overhead. For simple schemas, post-hoc repair *approaches* the validity of constrained decoding while remaining backend-agnostic. Complex schemas reveal the boundary. The 73.3% recovery rate drops at three or more nesting levels. Inter-field constraints exceed the capacity of two-layer repair.

Schema complexity as a repair failure mode predictor. The complex schema experiment (Table 3) shows a direct relationship. It predicts a clear relationship between schema depth and repair loop engagement. 83% of queries required at least one repair attempt. The 26.7% could not be recovered at all. Schema complexity is therefore a key predictor of post-hoc repair effectiveness. Future work should define a *complexity threshold*. Candidate metrics include nesting depth, required field count, and inter-field constraint density.

Retrieval grounding as a trade-off. Both retrieval conditions improve validity. They both provide grounding context but consistently reduce semantic alignment. Retrieval must be tuned for *semantic precision*—not just recall. Precision here means retrieved content that focuses rather than broadens generation context. TF-IDF degradation ($\Delta = -0.039$) exceeds embedding-based degradation ($\Delta = -0.006$). Dense retrieval provides better semantic focus than lexical retrieval in this setting.

Taken together, these three observations converge on a unified design principle for structured synthetic data pipelines: schema complexity and retrieval precision are the two primary levers governing the reliability-alignment trade-off, and both must be explicitly managed rather than treated as fixed background conditions.

Limitations. This study has four principal limitations. First, the standard schema benchmark uses $N = 100$ queries across a single schema type; generalization to diverse schema families requires additional evaluation. Second, Stage 2 uses $N = 15$ matched pairs, which is insufficient for statistical significance testing—these results should be treated as a pilot measurement. Third, no direct comparison is made against constrained decoding systems on the same queries. This is the most important gap. We are actively designing this comparison as the primary next experiment, targeting Outlines and Guidance on identical query sets with matched hardware constraints. Fourth, the repair loop convergence rate is characterized empirically but not formally bounded; a theoretical analysis of convergence conditions would strengthen the reliability guarantees.

7 Conclusion

In this work, we presented a controlled empirical study of structured synthetic data generation reliability using a fully local, consumer-grade pipeline. Our three main findings are: (1) multi-layer post-hoc schema enforcement raises JSON validity by 24–28 percentage points without model fine-tuning; (2) complex nested schemas trigger the repair loop on 83% of queries, with a clear two-layer failure taxonomy and a 73.3% recovery rate; and (3) both retrieval grounding and VLM conditioning introduce consistent but modest semantic alignment degradation. Together, these findings provide an empirical foundation for understanding the reliability-flexibility trade-off in post-hoc structured generation and motivate future

work on schema complexity thresholds and formal repair convergence bounds.

The pipeline and all experimental scripts are publicly available at https://github.com/ImSachin010/structured_synthetic_data_generation as supplementary material.

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